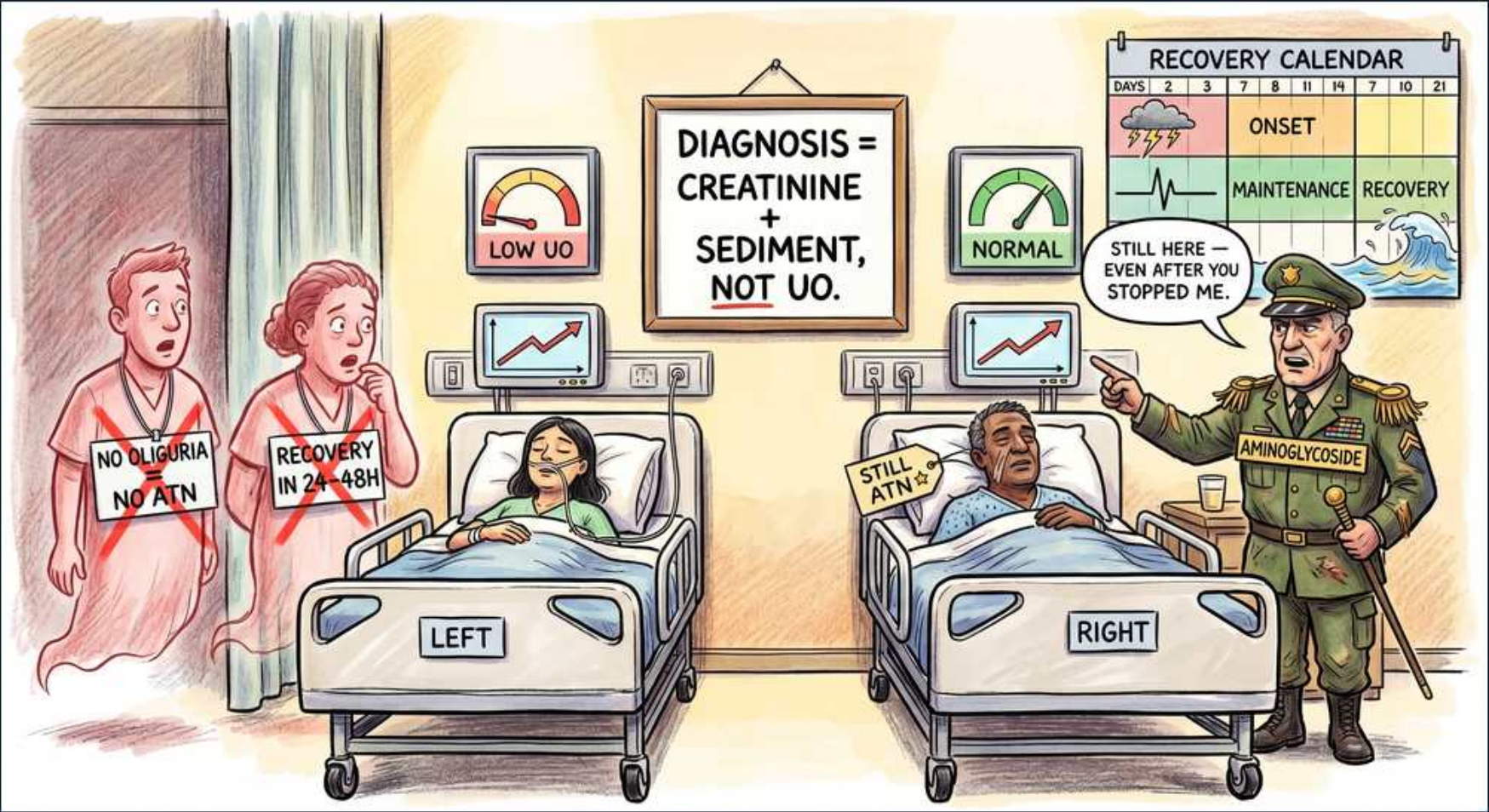


ATN: Diagnosis by Sediment, Not Urine Output



1. Dx = creatinine + sediment, NOT UOP

WHAT YOU SEE

Central sign 'DIAGNOSIS = CREATININE + SEDIMENT, NOT UO'

WHAT IT ENCODES

AKI/ATN is diagnosed by rising creatinine plus the urine sediment (muddy brown granular casts, RTE cells) — NOT by urine output. Both oliguric and non-oliguric ATN exist; non-oliguric carries a better prognosis.

BOARD TRAP

Excluding ATN because urine output is normal — non-oliguric ATN is common; rely on creatinine trend and sediment.

2. No oliguria ≠ no ATN

WHAT YOU SEE

Crossed-out patient 'NO OLIGURIA → NO ATN'

WHAT IT ENCODES

Maintained urine output does NOT rule out tubular necrosis. Non-oliguric ATN (UOP >400–500 mL/day) is frequent with nephrotoxins (aminoglycosides, contrast) and has lower mortality and less need for dialysis.

BOARD TRAP

Calling preserved UOP 'prerenal' or 'not AKI' — the sediment and creatinine define ATN regardless of output.

3. Recovery in 24–48h (not always)

WHAT YOU SEE

Crossed-out patient 'RECOVERY IN 24–48H'

WHAT IT ENCODES

Prerenal azotemia reverses within 24–48h of restoring volume; ATN does NOT — it persists for days to weeks. Failure to improve after a fluid challenge points away from prerenal toward established tubular injury.

BOARD TRAP

Expecting ATN to bounce back in 1–2 days like prerenal — ATN follows a prolonged maintenance phase before recovery.

4. Left bed: low UO but normal sediment

WHAT YOU SEE

Left monitor 'LOW UO'

WHAT IT ENCODES

Low urine output alone, with a bland sediment and a high BUN/Cr ratio, suggests prerenal physiology, not ATN. UOP is a poor discriminator without the sediment context.

BOARD TRAP

Diagnosing ATN from oliguria alone — a bland sediment with low UOP is more consistent with prerenal azotemia.

5. Right bed: 'still ATN' with normal UO

WHAT YOU SEE

Right patient badge 'STILL ATN', monitor 'NORMAL'

WHAT IT ENCODES

A patient with normal urine output can still have ATN — muddy brown casts and a rising creatinine confirm injury. This is the non-oliguric ATN pattern, classically from aminoglycosides.

BOARD TRAP

Reassuring on 'normal' urine output — normal UOP with rising Cr and granular casts is non-oliguric ATN.

6. Aminoglycoside — delayed, non-oliguric

WHAT YOU SEE

'AMINOGLYCOSIDE' general

WHAT IT ENCODES

Aminoglycosides cause classic non-oliguric ATN via proximal tubular accumulation. Injury is dose/duration related, often appears 5–10 days in, and Cr keeps rising even after the drug is stopped. Also causes hypomagnesemia/hypokalemia.

BOARD TRAP

Assuming the kidney recovers as soon as the drug is stopped — aminoglycoside ATN worsens for days after discontinuation.

7. 'Still here even after you stopped me'

WHAT YOU SEE

Speech bubble from the general

WHAT IT ENCODES

Aminoglycoside nephrotoxicity has a delayed, persistent course: tubular cells accumulated the drug and continue to die after cessation, so creatinine peaks late. Recovery follows once tubules regenerate.

BOARD TRAP

Misattributing the late Cr rise to a new insult — it is the expected lag of aminoglycoside tubular toxicity.

8. Recovery calendar: onset phase (days 2–3)

WHAT YOU SEE

'RECOVERY CALENDAR — ONSET' storm icons

WHAT IT ENCODES

ATN evolves through phases: initiation/onset (the insult, days 1–3), then maintenance. The onset phase is when GFR falls and the diagnosis is made; intervention here is limited to removing the insult and supportive care.

BOARD TRAP

Expecting reversal during the onset phase — damage is already done; the goal is preventing further injury.

9. Maintenance phase (days ~7–14)

WHAT YOU SEE

Calendar 'MAINTENANCE' segment

WHAT IT ENCODES

The maintenance phase is the prolonged plateau of low GFR (often 1–2 weeks) where complications occur: hyperkalemia, acidosis, volume overload, uremia. Most dialysis needs arise here. UOP may be low or normal.

BOARD TRAP

Mistaking the maintenance plateau for treatment failure — it is the expected protracted course of ATN before recovery.

10. Recovery / polyuric phase

WHAT YOU SEE

Calendar 'RECOVERY' wave icon

WHAT IT ENCODES

Recovery brings tubular regeneration and a polyuric phase (tubules regain GFR before reabsorptive capacity), risking volume depletion and hypokalemia/hypomagnesemia. Creatinine then trends down toward baseline.

BOARD TRAP

Overlooking electrolyte losses in the polyuric recovery phase — patients can become hypokalemic and volume-depleted.

11. Normal monitor framing

WHAT YOU SEE

'NORMAL' gauge over right bed

WHAT IT ENCODES

A 'normal' vital/output display reinforces that ATN can present without obvious red flags. Diagnosis rests on the laboratory creatinine trajectory and microscopic sediment, which the bedside numbers may not reveal.

BOARD TRAP

Trusting a normal-appearing bedside picture — order a creatinine and urine micro to catch non-oliguric ATN.

12. Rising creatinine trend graphs

WHAT YOU SEE

Upward-trending graphs on both monitors

WHAT IT ENCODES

Serial creatinine rise (KDIGO: $\uparrow \geq 0.3$ mg/dL in 48h or ≥ 1.5 x baseline) defines AKI. Trending Cr — not a single value — establishes the diagnosis and stages severity alongside urine output criteria.

BOARD TRAP

Using one creatinine value — AKI staging requires the change over time; a single 'normal-ish' Cr can miss early injury.
